

Cosmology

Exercises Sheet 2

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Exercise 1

To find the present density parameter of massive neutrinos (assuming direct hierarchy, and consider both relativistic and non-relativistic neutrinos) we can assume two separate cases: massless (aka relativistic) and massive (aka non-relativistic).

Let's start with the relativistic case. The density parameter is defined as:

$$\Omega_\nu = \frac{\rho_\nu}{\rho_{crit}} \quad (1)$$

with $\rho_{crit} = \frac{3H^2}{8\pi G}$.

For relativistic particles the energy density is:

$$\rho c^2 = \left[\frac{7}{8} \right] \frac{\pi^2}{30} \frac{k_B^4}{\hbar c^3} g T^4 \quad (2)$$

where $[7/8]$ it's true only for fermions. Indeed, if neutrinos are massless, they preserve their initial Fermi-Dirac distribution, although there are some deviations at low redshift, which will be discussed in the second part of the exercise.

To find the energy density of neutrinos (fermions), we can start from the energy density of photons (bosons), which is a well-known quantity. We want to relate the energy density of neutrinos to that of photons, that is:

$$\frac{\rho_\nu}{\rho_\gamma} = \frac{7}{8} \cdot \frac{g_\nu}{g_\gamma} \left(\frac{T_\nu}{T_\gamma} \right)^4 \quad (3)$$

The statistical weight are respectively $g_\nu = 3 \cdot 2$ (3 neutrinos generations for one spin state and 3 anti-neutrinos) and $g_\gamma = 2$ (2 spin states). So, $g_\nu/g_\gamma = 3$. Now let's see the temperature relationship. It's possible to demonstrate that the temperature ratio between neutrinos and photons (see Exercise 4) is:

$$\frac{T_\nu}{T_\gamma} = \left(\frac{4}{11} \right)^{\frac{1}{3}}. \quad (4)$$

Putting all these factors together leads to:

$$\rho_\nu = 3 \cdot \frac{7}{8} \cdot \left(\frac{4}{11}\right)^{\frac{4}{3}} \rho_\gamma \quad (5)$$

Knowing the today photons temperature $T_{0\gamma} = 2.7$ K, the energy density today is

$$\rho_{0\gamma} = \frac{\pi^2}{15} \frac{k_B^4}{\hbar^3 c^5} T_{0\gamma}^4 = 4,8 \cdot 10^{-34} \text{ g} \cdot \text{cm}^{-3} \quad (6)$$

Therefore, the energy density of relativistic neutrinos today is

$$\rho_{0\nu} = 3,27 \cdot 10^{-34} \text{ g} \cdot \text{cm}^{-3} \quad (7)$$

Using Eq. (1) we get the density parameter today for relativistic case:

$$\boxed{\Omega_{0\nu} = \frac{\rho_\nu}{\rho_{crit}} \Big|_{today} = \frac{1,7 \cdot 10^{-5}}{h^2}} \quad (8)$$

where we used that $\rho_{0crit} = 1,88 \cdot 10^{-29} h^2 \text{ g} \cdot \text{cm}^{-3}$ with $H_0 = 2,18 \cdot 10^{-18} \text{ Hz}$ ($= 67 \text{ km/s/Mpc}$) and we used the definition $h \equiv H/(100 \text{ km/s/Mpc})$, so today $h = 0.67$. In detail the numerical calculation is

$$\Omega_{0\nu} = \frac{\rho_{0\nu}}{\rho_{0crit}} = \rho_{0\nu} \frac{8\pi G}{3(100h \frac{km}{s^{-1}Mpc^{-1}})^2} = \rho_{0\nu} \frac{8\pi 6,67 \cdot 10^{-11} \frac{Nm^2}{kg^2}}{30000 \left(\frac{2,18 \cdot 10^{-18}}{67} \frac{1}{s}\right)^2 h^2} = \frac{3,27 \cdot 10^{-34} \text{ g} \cdot \text{cm}^{-3}}{1,88 \cdot 10^{-29} h^2 \text{ g} \cdot \text{cm}^{-3}} \cdot \quad (9)$$

Let us now move on to the non-relativistic case. At late times the massive neutrino are non relativistic, so the energy density is $\rho_\nu = n_\nu m_\nu^1$, so if we are interested in finding ρ_ν we must find n_ν . We know that neutrinos decouple at $T_\nu^{dec} \approx 0.8 \text{ MeV}$. After decoupling, the neutrinos move freely along geodesics and preserve to an excellent approximate the relativistic Fermi-Dirac distribution (even after they become non-relativistic at later times). While a massive neutrino would likely be non-relativistic today, we have assumed that their mass is low enough that they were relativistic when they decoupled from the thermal bath. Out of equilibrium, we need not do any integrals to determine their number density, because the neutrinos simply propagate along geodesics of spacetime. What matters is the initial condition. Therefore, at decoupling the neutrinos had a number density:

$$n_\nu = \frac{3}{4} \frac{\zeta(3)}{\pi^2} \left(\frac{k_b}{\hbar c}\right)^3 g_\nu T_\nu^3 \quad (10)$$

The photon temperature today, however, is given by relativistic thermodynamics, and conveniently enough has a simple form:

$$n_\gamma = \frac{\zeta(3)}{\pi^2} \left(\frac{k_b}{\hbar c}\right)^3 g_\gamma T_\gamma^3 \quad (11)$$

¹In reality the energy density in the non relativistic case is $\rho_\nu = n_\nu m_\nu + \frac{3}{2} k_B T$ but we can neglect the last term because at low (non-relativistic) speeds, the kinetic energy associated with thermal motion is negligible compared to the energy associated with mass.

Hence we can easily compute the ratio of the two number densities:

$$\frac{n_\nu}{n_\gamma} = \frac{3 g_\nu}{4 g_\gamma} \left(\frac{T_\nu}{T_\gamma} \right)^3 = \frac{3}{4} \cdot \frac{2}{2} \cdot \frac{4}{11} = \frac{3}{11} \quad (12)$$

where we have taken $g_\nu = 2$ because we are considering the number density of one generation of neutrinos and anti-neutrinos in the universe. In other words, we are doing as before (see Eq. (3)), but now with number densities.

We can therefore express the energy density as:

$$\rho_\nu = n_\nu m_\nu = \frac{3}{11} m_\nu n_\gamma. \quad (13)$$

We used this form because n_γ depends on the CMB temperature, a known quantity. Indeed, the number density of CMB photons today is

$$n_{0\gamma} = 410 \text{ cm}^{-3} \quad (14)$$

Using Eq. (1) we get the density parameter today for non relativistic case:

$$\boxed{\Omega_{0\nu} = \frac{\rho_\nu}{\rho_{crit}} \Big|_{today} = \frac{3}{11} \frac{m_\nu n_{0\gamma}}{\rho_{0crit}} = \frac{m_\nu}{94 h^2 \text{eV}}} \quad (15)$$

where we used that $\rho_{0crit} = 1,88 \cdot 10^{-29} \text{ h}^2 \text{ g} \cdot \text{cm}^{-3}$ as before and remembering that $1 \text{ kg} = 5,61 \cdot 10^{35} \text{ eV}$.

Exercise 2

Let us consider two frames K and K' rotated by a certain angle θ with a relative uniform velocity v along the x-axis. If a pulse of light is emitted at the origin at $t = 0$, each observer will see an expanding sphere centered on his own origin. This is a consequence of second postulate of relativity and is inconsistent with classical concepts, which would have the sphere always centered on a point at rest. The reconciliation of this result requires us to view both space and time as quantities peculiar to each observer and not universal.

It's possible to show that any periodic phenomenon in the moving frame K' will appear to have a longer period by a factor γ when viewed by local observers in frame K . If, on the other hand, we measure the arrival times of pulses or other indications of the periodic phenomenon that propagate with the velocity of light, then there will be an additional effect on the observed period due to the delay times for light propagation. The joint effect is called the *Doppler relativistic effect*.

In the rest frame of the observer K imagine that the moving source emits one period of radiation as it moves from point 1 to point 2 at velocity v . We also assume that the observer is located infinity. If the angular frequency of the radiation in the rest frame of the source (i.e K') is ω' then the time taken to move from point 1 to point 2 in K' is $\Delta t' = (2\pi)/\omega'$. While in the

observer's frame (i.e. K), time is given by the time dilation effect²:

$$\Delta t = \frac{2\pi}{\omega'} \gamma \quad (16)$$

Given l the distance between points 1 and 2, the first photon that the observer receives has traveled more space than the photon received at point 2 (having traveled an additional distance d). Figure (1) should clarify the dynamics.

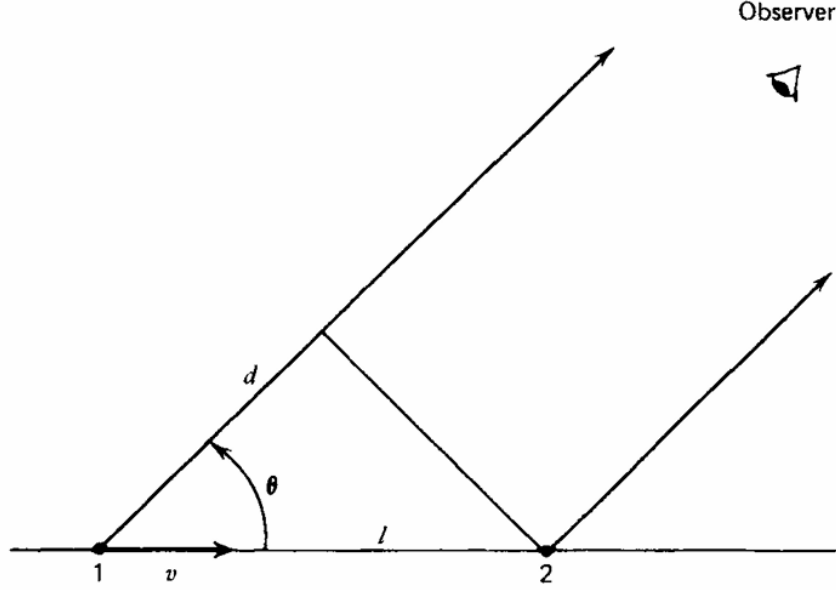


Figure 1: Geometry for the Doppler effect.

We have called d the distance between 1 and the right triangle generated by the rays of light arriving at the observer from 1 and 2. From Figure (1) it is possible to notice that $l = v\Delta t$ and $d = v\Delta t \cos(\theta)$.

If the frame K' wasn't inclined by an angle, there would simply be time dilation and both photons emitted by the CMB, i.e. the first photon at a time t'_1 and the second photon at a time t'_2 , would have traveled the same distance. So the time difference, in the observer's frame, would only be $\Delta t_A = t_2 - t_1 = \Delta t' \gamma$. However, this is not the situation, in fact in our case the first photon has traveled a distance d more than the second photon. This translates into a longer time for the first photon, i.e. $t_1 + d/c$. While for the second photon it remains t_2 . Consequently:

$$\Delta t_A = t_2 - \left(t_1 + \frac{d}{c}\right) = \Delta t - \frac{d}{c} = \Delta t \left(1 - \frac{v}{c} \cos(\theta)\right) \quad (17)$$

Hence in term of angular frequency is:

$$\omega = \frac{2\pi}{\Delta t_A} = \frac{2\pi}{\Delta t(1 - \frac{v}{c} \cos(\theta))} = \frac{\omega'}{\gamma(1 - \frac{v}{c} \cos(\theta))} \quad (18)$$

²We remember that the time dilation formula is: $\Delta t = \gamma \Delta t'$ where $\Delta t' = t'_2 - t'_1$ is the proper time, i.e. the time interval that elapses between two instants and which is measured at the same point of the reference system by a clock integral with the observer, i.e. K' .

From a physical point of view, the γ factor is the relativistic correction factor due to time dilation and $(1 - \frac{v}{c} \cos(\theta))$ is the correction factor due to classical Doppler. The relationship that connects frequency to temperature in blackbody spectrum is the Wien's displacement law:

$$\lambda_{peak} = \frac{c}{\nu_{peak}} = c \frac{2\pi}{\omega_{peak}} = \frac{b}{T} \quad (19)$$

where T is the temperature and b is a constant of proportionality called Wien's displacement constant. Then it is clear that a shift in frequency corresponds to a shift in temperature, so (18) is also valid for temperature:

$$T = \frac{T'}{\gamma(1 - \beta \cos(\theta))} \quad (20)$$

We note that the blackbody spectrum will remain the same because:

$$\frac{B_\nu}{\nu^3} \text{ IS LORENTZ INVARIANT} \quad (21)$$

Exercise 3

To show the maximum possible value of the primordial helium fraction, we consider that BBN proceeds until every free neutron is bonded into an atomic nucleus. Thus, suppose that after the proton-neutron freezeout, all neutrons are processed into helium, that is ${}^4He = 2n + 2p$. Since two neutrons go into 4He , the final 4He abundance is equal to half the neutron abundance. In term of mass fraction, the mass fraction of helium 4He is defined as

$$X_{He} = Y = \frac{m_{He}}{m_{He} + m_H} \quad (22)$$

Now, we assume $m_p = m_n$. Since one 4He nucleus has a mass of $4 \cdot m_p$, where m_p is the mass of proton, n_{He} such nuclei will have a mass of $(n_{He}) \cdot 4m_p$. Instead n_H nuclei of hydrogen have a mass of $(n_H) \cdot m_p$. Therefore, we can write the mass ratio as number density ratio:

$$Y = \frac{4n_{He}}{4n_{He} + n_H} \quad (23)$$

Considering the relationship between the number of $\{n_{He}, n_H\}$ and $\{n_p, n_n\}$, as said before, helium contains 2 protons, and hydrogen contains 1 proton, so $n_p = n_H + 2n_{He}$. While, the neutrons are only contributed to by helium in the accounting $n_n = 2n_{He}$. Inverting these relations yields

$$\begin{aligned} n_H &= n_p - n_n \\ n_{He} &= n_n/2 \end{aligned} \quad (24)$$

Here it can be noted that the number density of protons and neutrons is not the same because of the neutron-to-proton ratio, that is $\frac{n_n}{n_p} = \frac{1}{7}$. Therefore, the maximum value of mass fraction for the assumption that all neutrons are processed into helium is

$$Y_{max} = \frac{2n_n}{2n_n + (n_p - n_n)} = \frac{2n_n}{n_n + n_p} = \frac{2(n_n/n_p)}{1 + (n_n/n_p)} = \frac{2f}{1 + f} \quad (25)$$

where $f = n_n/n_p$.

Exercise 4

a)

The relativistic species (ν) leave the thermal bath with a distribution function reflecting what they were at the time of decoupling, so they are still described by relativistic equation:

$$n = \frac{g}{(2\pi\hbar)^3} \int \frac{1}{e^{\frac{pc}{k_B T}} + 1} 4\pi p^2 dp \quad (26)$$

where we have neglected the chemical potential because in the early universe $\mu \ll T$ and the mass because $m \ll T$ in relativistic contest. Indeed, since neutrinos are no longer interacting with any other particles, no particle collision can change this distribution from that point onward. Therefore after decoupling, the number of neutrinos in a comoving volume does not change so that Equation (26) is still valid. The number of neutrinos is given by:

$$N_\nu = n_\nu R^3 = \text{const} \quad (27)$$

This particle number conservation requires that $n_\nu \propto R^{-3}$, this means $n_\nu \propto a^{-3}$ remembering that $a = R/R_0$. It is therefore convenient to define the time-independent combination $q = ap$, so that the neutrino number density is

$$n_\nu = a^{-3} \frac{g_\nu}{(2\pi\hbar)^3} \int \frac{1}{e^{\frac{qc}{k_B a T_\nu}} + 1} 4\pi q^2 dq \quad (28)$$

This is only consistent with particle number conservation if $T_\nu \propto a^{-1}$. Hence, the temperature must decrease inversely proportional to the scale factor a .

b)

To find the effective temperature of the relic background of relativistic neutrinos today, we can use the conservation of entropy. The second law of thermodynamics states that the entropy S is always either constant or increasing $dS \geq 0$. If the Universe is in thermal equilibrium as it expands, it turns out that the inequality is saturated and the entropy of the Universe is constant. In thermodynamics language, this means that we can consider the expansion of the Universe as an adiabatic (reversible) process. To see this, we first need to derive an expression for the entropy of the Universe in terms of quantity we already know like the energy density ρ and the pressure P . To do so, we will need the following relation:

$$\frac{\partial P}{\partial T} = \frac{\rho c^2 + P}{T} \quad (29)$$

which is valid in thermal equilibrium with $\frac{\mu}{T} \ll 1$. Now, from the first law of thermodynamics

$$dS = \frac{1}{T} [dU + PdV]. \quad (30)$$

Writing $U = \rho c^2 R^3$, and $V = R^3$ we get:

$$\begin{aligned}
dS &= \frac{1}{T} [d(R^3 \rho c^2) + P d(R^3)] \\
&= \frac{1}{T} [d(R^3(\rho c^2 + P)) - R^3 dP] \\
&= \frac{1}{T} \left[d(R^3(\rho c^2 + P)) - \frac{R^3}{T} (\rho c^2 + P) dT \right] \\
&= \frac{1}{T} d[R^3(\rho c^2 + P)] - \frac{R^3}{T^2} (\rho c^2 + P) dT \\
&= d \left[\frac{R^3(\rho c^2 + P)}{T} + \text{const} \right]
\end{aligned} \tag{31}$$

which shows that the entropy of the Universe is simply:

$$S = R^3 \frac{\rho c^2 + P}{T} = \text{const} \tag{32}$$

This is the entropy of the entire universe. Now that we know the definition of entropy, we can compute its value. The value is always completely dominated by relativistic particles (i.e. radiation) since non-relativistic particles in our Universe have either exponentially suppressed abundances ($n \propto e^{-\frac{mc^2}{T}}$), or have number densities that are orders of magnitude smaller than that of photons. Thus, for radiation, we have $P = \rho/3$, so S is given by:

$$S = \frac{R^3}{T} \rho_{rel} c^2 \cdot \frac{4}{3} \tag{33}$$

The expression for $\rho_{rel} c^2$ is:

$$\rho_{rel} c^2 = \left[\frac{7}{8} \right] \frac{\pi^2}{30} \frac{k_B^4}{\hbar^3 c^3} g_* T^4 \tag{34}$$

where $[7/8]$ applies only to fermions, thus the entropy takes the following form:

$$S = \frac{4 \pi^2}{3} \frac{k_B^4}{30 \hbar^3 c^3} R^3 \left(\sum_{i \in \text{bosons}} g_i T_i^3 + \frac{7}{8} \sum_{j \in \text{fermions}} g_j T_j^3 \right) = \frac{2 \pi^2}{45} \frac{k_B^4}{\hbar^3 c^3} g_{*S}(T) (RT)^3 \tag{35}$$

where g_{*S} is the *effective number of degrees of freedom in entropy*. We see that if all particle in the Universe are in thermal equilibrium at the same temperature, we have $g_{*S} = g_*$. This is true before neutrino decoupling, but not after. Conservation of entropy implies that:

$$T \propto g_{*S}^{-\frac{1}{3}} R^{-1} \tag{36}$$

When g_{*S} is constant, we retrieve the standard result that $T \propto 1/a$. However, when g_{*S} is changing T no longer decays as $1/a$, but instead decays slower. This reflects the fact that when g_{*S} changes, particles are disappearing from the plasma and transferring their entropy and energy to other particles in the plasma, causing their temperature to decay less fast. We are now

ready to use entropy conservation to compute the temperature ratio $\frac{T_\nu}{T_\gamma}$. First, let's consider the entropy density of the Universe at time t_1 (scale factor $a_1 = a(t_1)$ and temperature T) before annihilation. Before annihilation, the fermions were electrons (2 spin states), positrons (2), neutrinos (3 generations and one spin state) and anti-neutrinos (3). The bosons are photons (2 spin states). So at a_1 the entropy is

$$S(a_1) = \frac{2\pi^2}{45} [2 + (7/8)(2 + 2 + 3 + 3)] T^3 = \frac{43\pi^2}{90} T^3 \quad (37)$$

where $T = T_\nu = T_\gamma$. After electrons and positrons annihilation the entropy density at time t_2 (scale factor $a_2 = a(t_2)$) is

$$S(a_2) = \frac{2\pi^2}{45} \left[2T_\gamma^3 + \frac{7}{8}6T_\nu^3 \right] \quad (38)$$

where we have now allowed for photons and neutrinos to have different temperatures since photons get the entropy and energy from the annihilating e+e- while neutrinos do not.

Since the entropy is constant we have $S(a_1) = S(a_2)$:

$$\frac{43}{2} T^3 = 2 \left[2T_\gamma^3 + \frac{7}{8}6T_\nu^3 \right] = 4 \left[\left(\frac{T_\gamma}{T_\nu} \right)^3 + \frac{21}{8} \right] T_\nu^3 \quad (39)$$

But as said before $T = T_\nu$. Therefore, the ratio of the two temperatures is

$$\frac{T_\nu}{T_\gamma} = \left(\frac{4}{11} \right)^{\frac{1}{3}}. \quad (40)$$

Finally, since the cosmic photon background temperature today has cooled to $T_{0\gamma} = 2.73K$, the effective temperature of the relic background of relativistic neutrinos today is

$$T_{0\nu} = \left(\frac{4}{11} \right)^{\frac{1}{3}} T_{0\gamma} = 1.95K. \quad (41)$$